

Real-Time SCADA Intrusion Detection Under Communication Constraints: Latency, Payload, and Generalization in Industrial Edge Environments

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Abstract—Digital substations increasingly rely on continuous communication among sensing, protection, and supervisory components to support timely operational decisions. Intrusion detectors within this workflow must respond quickly, operate with the telemetry visible at their processing location, and remain reliable under conditions not observed during development. However, SCADA intrusion detection is commonly evaluated as an offline classification task, separating model accuracy from communication payload, processing latency, telemetry availability, and cross-run transfer.

We examine this mismatch through a communication-aware evaluation of four lightweight detectors on the MSU/ORNL Power System Attack Dataset. Using fifteen-fold leave-one-run-out testing, we study unseen-run discrimination, batch-size-one pipeline latency, modeled payload, missing telemetry, and edge-visible versus central feature conditions. Random Forest ROC-AUC decreases from 0.972 under random splitting to 0.677 under run-aware evaluation. Logistic Regression, XGBoost, and a compact multilayer perceptron meet the 10 ms and 16.7 ms P95 study references, whereas Random Forest meets only the 100 ms reference. A training-fold-selected 16-feature condition reduces modeled payload from 512 to 64 bytes while preserving comparable or higher mean ROC-AUC for three models. Central telemetry improves clean-condition discrimination, but relay-associated feature loss narrows or removes this advantage.

These findings show that low-latency inference alone does not establish the usefulness of a real-time industrial detector. Credible evaluation must consider unseen-run transfer, telemetry efficiency, placement-dependent visibility, degradation resilience, and false-alarm burden as connected properties of the communication workflow.

Index Terms—real-time industrial communications, SCADA security, intrusion detection, edge computing, industrial IoT, telemetry constraints, digital substations, machine learning

I. INTRODUCTION

Digital substations depend on time-sensitive communication among phasor measurement units (PMUs), protective relays, control systems, and security monitors. An intrusion detector in this workflow must not only classify telemetry accurately, but also operate on the measurements available at its location, meet practical processing budgets, and avoid excessive false alarms.

These requirements create a real-time communication trade-off. Edge detection can reduce delay and communication dependence, but local models may have limited visibility. Central detection offers broader telemetry, but depends on timely delivery and may degrade under missing measurements or relay-channel loss. Reducing features can lower payload and processing cost, but may also remove useful attack evidence.

SCADA intrusion detection is nevertheless often evaluated as an offline classification task. Random row-level splitting can place records from the same collection run in both training and test sets, overstating performance under unseen conditions. Latency, payload, telemetry loss, and detector placement are also commonly evaluated separately from detection quality.

We therefore frame SCADA intrusion detection as a *communication-budgeted service*. Using fifteen-fold leave-one-run-out testing, we evaluate Logistic Regression, Random Forest, XGBoost, and a compact multilayer perceptron across unseen-run discrimination, batch-size-one latency, modeled payload, telemetry loss, and edge-visible versus central feature conditions. These placement conditions are analytical proxies rather than a deployed substation architecture.

A. Research Questions

- **RQ1:** Which detector pipelines meet the study’s latency references?
- **RQ2:** How does feature reduction affect payload, latency, and unseen-run discrimination?
- **RQ3:** How does telemetry loss affect detector performance?
- **RQ4:** When does central telemetry outperform edge-visible telemetry?

B. Contributions

This paper contributes:

- A communication-aware evaluation of SCADA intrusion detection across latency, payload, telemetry availability, and placement.

- A leakage-controlled, run-aware protocol in which all preprocessing, feature selection, training, and threshold selection exclude the held-out run.
- Evidence that reduced telemetry can preserve competitive unseen-run performance, while low latency and central visibility alone do not guarantee robust detection.

II. RELATED WORK

A. SCADA Intrusion Detection Beyond Classification Accuracy

Power-system intrusion detection has been studied using physical telemetry, cyber logs, and hybrid feature sets. The MSU/ORNL benchmark has supported disturbance and cyber-attack discrimination with synchronized measurements [2]–[4]. Later work has examined ensemble learning, feature selection, deep models, federated learning, and lightweight detection [7]–[10]. Surveys further show that dataset design, class balance, and evaluation protocols strongly shape reported performance [5], [6]. Rather than proposing a new classifier, we study whether lightweight detectors remain useful on unseen runs and under constrained telemetry.

B. Communication Constraints and Evaluation Design

IEC 61850 defines communication requirements for power-utility automation, while IEEE C37.118.2 specifies synchrophasor data transfer [12]–[14]. These standards establish the real-time communication context of this work, although our measurements do not demonstrate protocol compliance or protection-loop suitability. Low computational cost alone is insufficient when evaluation mixes collection conditions or assumes complete telemetry. Arp *et al.* identify leakage and unrealistic assumptions as recurring weaknesses in machine-learning security research [11]. We therefore evaluate run-aware generalization together with latency, payload, telemetry degradation, and detector placement.

III. COMMUNICATION-BUDGETED DETECTION FRAMEWORK

Our framework follows the operational path shown in Fig. 1: telemetry is observed under processing, payload, and availability constraints before classification at an edge-visible or central location. It is guided by three goals. **G1: credible transfer** requires testing on a complete unseen collection run. **G2: visible resource trade-offs** connects feature count to payload and measured pipeline latency. **G3: explicit degradation** evaluates what remains useful when measurements or relay-associated channels are unavailable.

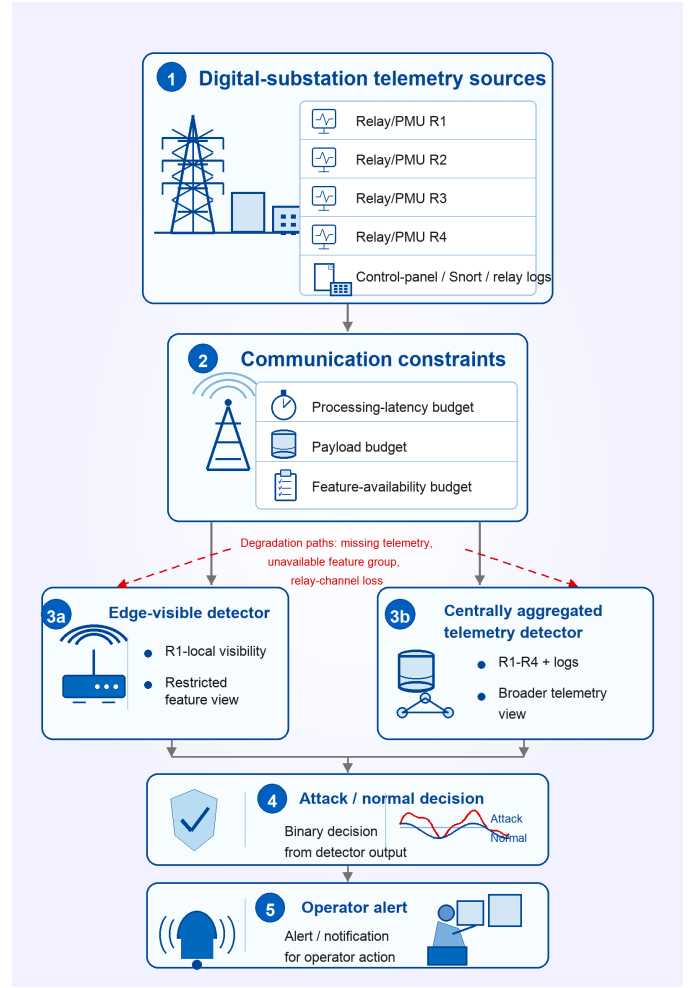


Fig. 1. Overview of the communication-budgeted evaluation framework. Digital-substation telemetry is examined under processing, payload, and availability constraints before classification with edge-visible R1 measurements or centrally aggregated features. The figure describes an analytical evaluation workflow, not a deployed production architecture.

A. Dataset and Visibility Conditions

We use the binary-labeled MSU/ORNL power-system attack dataset from the public Industrial Control System Cyber Attack Datasets collection [1]–[3]. The audited corpus contains fifteen runs and 78,377 records: 55,663 attack and 22,714 normal/natural records. Its 128 usable features comprise 116 PMU/relay measurements and 12 control-panel, Snort, or relay-log variables. Each file receives a `run_id` used only for grouped evaluation; it is never a model input.

The edge-visible condition uses the 29 R1 variables whose names begin R1– or R1::; models are retrained on these measurements only. The central condition uses all 128 features. The R1 result is one predefined proxy, not an average or best case across relays. Because the public files do not preserve a valid continuous event sequence, we evaluate record-level classification and do not claim attack-onset delay, early warning, or time-to-detection.

IV. METHOD

A. Run-Aware Evaluation and Models

Each leave-one-run-out fold trains on fourteen complete runs and tests on the remaining run. Imputation, scaling, feature ranking, model fitting, and threshold selection use training runs only; availability degradation is applied only to held-out records. Results are unweighted fold means and standard deviations unless noted. A stratified 70/30 random split with seed 42 is retained only as a contrast baseline.

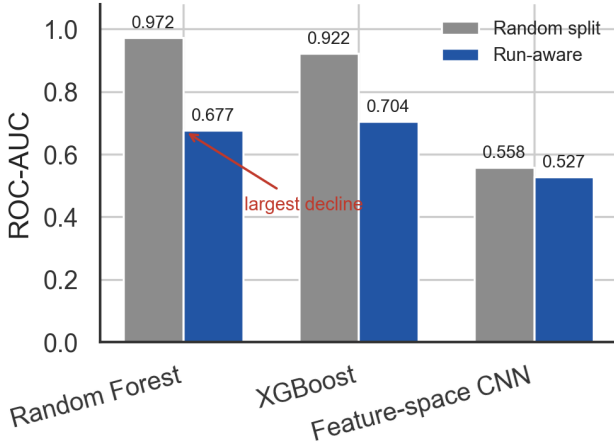


Fig. 2. Random row-level splitting versus leave-one-run-out evaluation for comparable baseline models. The figure highlights how discrimination changes when the test set is a complete unseen collection run.

All primary models use training-fitted median imputation. Logistic Regression and the compact MLP also use training-fitted standardization. Table I summarizes the configurations; all use seed 42.

TABLE I
PRIMARY MODEL CONFIGURATIONS.

Model	Main configuration
Logistic Regression	Standardized; <code>lbfgs</code> , ℓ_2 , $C = 1$, 2,000 maximum iterations.
Random Forest	200 Gini trees, square-root feature rule, bootstrap enabled, unrestricted depth.
XGBoost	200 estimators, depth 6, learning rate 0.3, histogram tree method, four CPU threads.
Compact MLP	Standardized; hidden layers (32, 16), ReLU, Adam, early stopping, 200 maximum iterations.

B. Processing and Payload Measurements

Batch-size-one latency is measured on cached fold-01 pipelines using 500 warm-up and 5,000 timed calls with `time.perf_counter_ns()`. Timing includes the fitted pre-processor and estimator `predict_proba()` call, but excludes acquisition, file I/O, serialization, transport, queueing, visualization, and operator response. Measurements were collected on an ARM64 macOS host with ten cores and 32 GB memory using Python 3.9.6, scikit-learn 1.6.1, and XGBoost 2.1.4. The

exact host model and power settings were not preserved, so results describe one recorded environment.

We report median and P95 latency. Median-equivalent rate is

$$R_{eq} = \frac{1000}{L_{median}}, \quad (1)$$

a derived single-record rate rather than measured concurrent throughput. P95 is compared with B1 (10 ms, strict application), B2 (16.7 ms, a 60-record/s interval), and B3 (100 ms, operational analytics). These study references do not represent complete sensor-to-application delay or standards compliance.

For n_f transmitted 32-bit numeric fields, modeled payload is

$$P = 4n_f \text{ bytes per record.} \quad (2)$$

Thus, 128, 64, 32, and 16 fields correspond to 512, 256, 128, and 64 bytes. The estimate excludes framing, timestamps, identifiers, security metadata, acknowledgments, retransmissions, and packetization.

Within each fold, a 100-tree Random Forest ranks features using only the fourteen training runs. The same fold-specific ranking is shared by all four models, which are retrained on the top 64, 32, or 16 features. For the 128-versus-16 comparison, paired fold differences are bootstrapped with 50,000 resamples and seed 20260713.

C. Telemetry Degradation and Threshold Transfer

Random missingness is applied independently to held-out feature cells at 5%, 10%, 20%, and 30% using seeds 42–46. Group loss sets selected test features to NaN, after which the training-fitted median imputer is applied. “Logs off” masks the 12 control-panel, relay-log, and Snort variables; “PMU off” masks all PMU/relay measurements. Relay loss masks the 29 measurements associated with R1, R2, R3, or R4. For each fold, the worst-relay value is the minimum ROC-AUC across those four stress cases.

For central 128-feature models, fixed 0.5 and training-selected thresholds (maximum F1, maximum Youden’s J , and training $FPR \leq 0.10$) transfer unchanged to the unseen run. Alert volume is predicted positives per 1,000 records.

V. RESULTS

A. Random Splits Overstate Unseen-Run Performance

TABLE II
BASELINE PERFORMANCE. RANDOM-SPLIT VALUES USE A 70/30 SPLIT; RUN-AWARE VALUES ARE MEAN $\pm$ STANDARD DEVIATION ACROSS FIFTEEN HELD-OUT RUNS.

Protocol	Model	Precision	Recall	F1	FPR	ROC-AUC
Random	Random Forest	0.915	0.976	0.945	0.224	0.972
Random	XGBoost	0.871	0.950	0.908	0.346	0.922
Random	Feature-space CNN	0.710	1.000	0.831	1.000	0.558
Run-aware	Logistic Regression	0.719 $\pm$ 0.043	0.976 $\pm$ 0.005	0.828 $\pm$ 0.029	0.934 $\pm$ 0.020	0.687 $\pm$ 0.031
Run-aware	Random Forest	0.760 $\pm$ 0.035	0.861 $\pm$ 0.012	0.807 $\pm$ 0.018	0.667 $\pm$ 0.028	0.677 $\pm$ 0.029
Run-aware	XGBoost	0.763 $\pm$ 0.038	0.853 $\pm$ 0.009	0.805 $\pm$ 0.022	0.650 $\pm$ 0.038	0.704 $\pm$ 0.028
Run-aware	Compact MLP	0.743 $\pm$ 0.037	0.904 $\pm$ 0.016	0.815 $\pm$ 0.024	0.767 $\pm$ 0.035	0.683 $\pm$ 0.022
Run-aware	Feature-space CNN	0.711 $\pm$ 0.043	0.998 $\pm$ 0.006	0.829 $\pm$ 0.029	0.998 $\pm$ 0.004	0.527 $\pm$ 0.038

Changing the evaluation boundary reduces Random Forest ROC-AUC from 0.972 to 0.677, a 0.295 drop. XGBoost has the highest mean run-aware ROC-AUC among the four primary

models. Logistic Regression and the diagnostic CNN retain favorable F1 values despite very high FPR, showing that F1 alone can hide weak unseen-run behavior.

B. Three Models Meet the Strict Processing References

TABLE III
BATCH-SIZE-ONE DETECTOR-PIPELINE LATENCY. CHECK MARKS INDICATE THAT P95 FALLS WITHIN THE CORRESPONDING STUDY REFERENCE.

Model	Median (ms)	P95 (ms)	Med.-eq. rec./s	B1	B2	B3
Logistic Regression	1.080	1.112	926	✓	✓	✓
Random Forest	15.669	51.711	64	—	—	✓
XGBoost	1.224	4.864	817	✓	✓	✓
Compact MLP	1.160	1.690	862	✓	✓	✓

Logistic Regression, XGBoost, and the compact MLP meet all three P95 references; Random Forest meets only B3. XGBoost therefore combines the best mean run-aware discrimination with compliance to the strict study references.

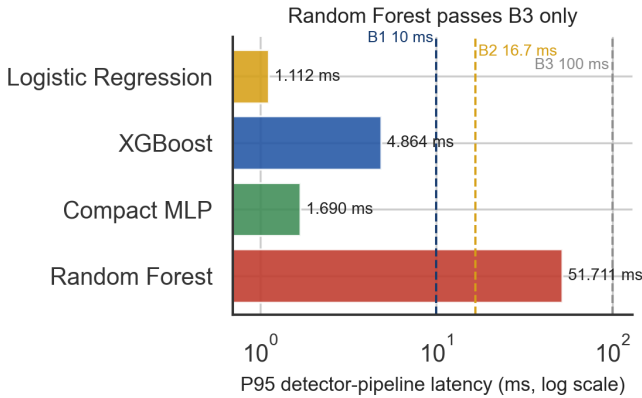


Fig. 3. Batch-size-one detector-pipeline P95 latency for the four primary models, shown against the 10 ms, 16.7 ms, and 100 ms study references.

C. Sixteen Features Can Preserve Detection While Cutting Payload

TABLE IV
RUN-AWARE RESULTS BY SHARED TRAINING-FOLD FEATURE BUDGET. PAYLOAD ASSUMES 32-BIT NUMERIC FIELDS AND EXCLUDES PROTOCOL OVERHEAD.

Model	Budget	Features	Payload (B)	ROC-AUC	P95 (ms)
Logistic Regression	Full	128	512	0.687	1.112
	Ranked	64	256	0.683	1.223
	Ranked	32	128	0.626	1.026
	Ranked	16	64	0.629	0.781
Random Forest	Full	128	512	0.677	51.711
	Ranked	64	256	0.674	28.624
	Ranked	32	128	0.672	15.992
	Ranked	16	64	0.721	15.353
XGBoost	Full	128	512	0.702	4.864
	Ranked	64	256	0.693	0.970
	Ranked	32	128	0.690	0.911
	Ranked	16	64	0.708	0.818
Compact MLP	Full	128	512	0.683	1.690
	Ranked	64	256	0.683	0.899
	Ranked	32	128	0.693	0.852
	Ranked	16	64	0.706	0.887

The 16-feature condition cuts modeled payload eightfold. Its paired 16-minus-128 ROC-AUC differences are -0.058 for Logistic Regression, $+0.045$ for Random Forest, $+0.006$ for XGBoost, and $+0.024$ for the compact MLP. Bootstrap 95% intervals are $[-0.071, -0.043]$, $[0.034, 0.055]$, $[-0.002, 0.016]$, and $[0.015, 0.033]$, respectively. Thus, reduction is model-dependent. For XGBoost, mean ROC-AUC remains comparable while P95 falls from 4.864 to 0.818 ms; the small AUC increase is not established by the interval.

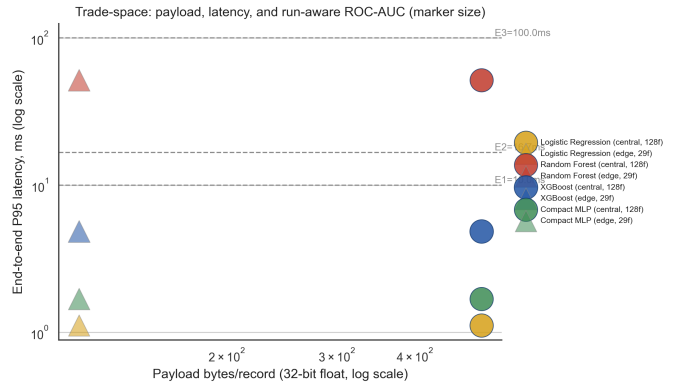


Fig. 4. Payload-latency trajectories from 128 features to the shared training-fold-selected 16-feature condition. The figure connects modeled payload, detector-pipeline P95 latency, and run-aware ROC-AUC; network and protocol overhead are excluded.

D. Physical Telemetry Dominates Under Degradation

TABLE V
RUN-AWARE ROC-AUC UNDER HELD-OUT-TEST AVAILABILITY DEGRADATION.

Model	Clean	10% Missing	30% Missing	Worst Relay	Logs Off	PMU Off
Logistic Regression	0.687	0.581	0.543	0.499	0.672	0.531
Random Forest	0.677	0.680	0.659	0.580	0.674	0.506
XGBoost	0.702	0.672	0.617	0.612	0.699	0.509
Compact MLP	0.683	0.636	0.589	0.564	0.659	0.518

Random Forest changes least under random missingness, while XGBoost retains the strongest fold-wise worst-relay ROC-AUC. Removing cyber/log variables has little effect, but removing all PMU/relay measurements reduces every model to near-chance discrimination. The evaluated detectors therefore rely primarily on physical telemetry.

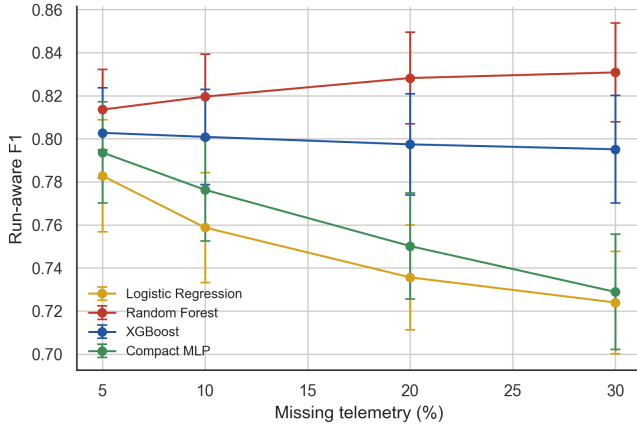


Fig. 5. Run-aware F1 under 5%, 10%, 20%, and 30% test-time random missingness. Markers show means over fifteen held-out runs and five masking seeds; error bars show standard deviation across saved fold-seed results.

E. Central Visibility Helps Until Relay Loss Narrows the Advantage

TABLE VI
MODEL-SPECIFIC ROC-AUC UNDER ANALYTICAL EDGE-VISIBLE AND CENTRAL TELEMETRY CONDITIONS.

Model	Edge R1	Central clean	Central 10% miss.	Logs off	Worst relay
Logistic Regression	0.531	0.687	0.581	0.672	0.499
Random Forest	0.564	0.677	0.680	0.674	0.580
XGBoost	0.579	0.702	0.672	0.699	0.612
Compact MLP	0.550	0.683	0.636	0.659	0.564

Central telemetry outperforms the R1 edge-visible condition for every model when clean, and remains stronger under 10% random missingness and log loss. Relay-associated loss narrows this advantage. Logistic Regression falls to 0.499 in its fold-wise worst-relay condition, below its edge-visible result of 0.531; XGBoost retains the strongest degraded-central result at 0.612.

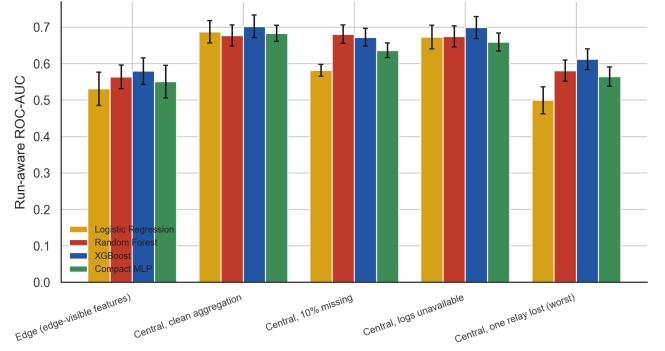


Fig. 6. Run-aware ROC-AUC for the edge-visible R1 condition and central telemetry under clean and degraded availability. These are analytical feature-visibility conditions rather than deployed systems.

F. Threshold Transfer Leaves a High Alert Burden

At the fixed 0.5 threshold, the four-model mean is 856 alerts per 1,000 records (FPR 0.755, recall 0.898). Training-based optimization reduces but does not solve the problem: the FPR-constrained rule still produces 524 alerts (FPR 0.357) while recall falls to 0.592. Cross-run score separation, not processing speed alone, remains the limiting factor.

VI. DISCUSSION

A. Fast Inference Is Necessary, but Not Sufficient

Three models meet the strict latency references, yet all retain substantial false-positive behavior on unseen runs. Speed therefore does not establish operational usefulness. Real-time evaluations should report latency alongside ROC-AUC, FPR, class balance, and alert volume.

B. Telemetry Reduction Depends on the Model

Reducing the feature set from 128 to 16 cuts modeled payload from 512 to 64 bytes, but the effect varies by model. Random Forest and the compact MLP improve, XGBoost remains comparable, and Logistic Regression declines. Telemetry reduction should therefore be selected within the training boundary and validated for the intended detector.

C. Central Visibility Increases Channel Dependence

Central telemetry improves clean-condition discrimination, but relay loss can narrow or remove this advantage. This trade-off motivates fallback models, redundant channels, and confidence-aware abstention, although we do not evaluate these mechanisms here.

D. The Best Trade-Off Is Not Deployment-Ready

The 16-feature XGBoost configuration provides the strongest measured balance: 64 modeled bytes, 0.708 ROC-AUC, and 0.818 ms P95 latency. However, its false-positive and threshold-transfer burden remains too high for deployment without further calibration and system-level testing.

VII. THREATS TO VALIDITY AND LIMITATIONS

The study uses one attack-majority benchmark from one testbed and cannot establish transfer across utilities, vendors, topologies, or attack taxonomies. The files lack a valid continuous event sequence, preventing claims about onset delay, queueing, temporal drift, or early warning.

Latency was measured with cached fold-01 pipelines in one recorded ARM64 macOS environment. The exact host model, CPU-affinity policy, and power mode were not preserved. Measurements include preprocessing and prediction but exclude acquisition, serialization, transport, switching, retransmission, visualization, and operator response. The study references therefore do not establish IEC 61850 compliance or protection-loop suitability.

Payload counts only assumed 32-bit numeric fields and excludes protocol overhead. Missingness and relay loss are feature-availability experiments, not packet-level replay. Feature ranking uses Random Forest impurity importance and may vary by fold. Edge and central conditions are analytical proxies, and the fold-wise worst-relay result is a stress case rather than an expected failure. Threshold results average four models; the diagnostic CNN is excluded from all communication-budget experiments.

VIII. CONCLUSION

We examined SCADA intrusion detection as a communication-budgeted service rather than an accuracy-only classification problem. Run-aware testing exposed a substantial performance gap hidden by random splitting, while favorable F1 scores sometimes concealed severe false-positive behavior. Three lightweight models met the strict processing references, and a 16-feature condition reduced modeled payload eightfold while preserving model-dependent unseen-run performance. Central telemetry improved clean-condition discrimination, but its advantage became less reliable under relay-associated feature loss.

These results suggest that lightweight inference is only one part of effective real-time industrial security. Future systems must jointly support unseen-run generalization, communication efficiency, placement-aware visibility, false-alarm control, and resilience to telemetry degradation.

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